

HIDDEN TREASURE The rocky hill in Salal village where the GSI has found lithium resources

surge in EV production with 223 per cent growth registered in 2022, accounting for 48,000 vehicles, according to analyst firm Canalys. The firm forecasts that the EV expansion in the country will hit over 300,000 units in 2025, representing over a 6 per cent share of the total light car market. But India's dependence on China for the import of Li-ion batteries is considered a major bump in the ambitious *aatmanirbhar* (self-reliant) initiative. This was pointed out by the Indian chapter of the World Resources Institute (WRI) in its 2022 report. It stated that the supply of minerals needed for commercially available battery technologies—lithium, cobalt and nickel—is dominated by a handful of countries; another roadblock for India. Rishabh Jain, senior programme lead, Council on Energy, Environment and Water (CEEW), says

LITHIUM IS WIDELY USED IN BATTERIES DUE TO ITS HIGH ELECTROCHEMICAL POTENTIAL AND LOW WEIGHT

85 per cent of lithium/Li-ion cells were imported from Australia, Chile and China in 2022.

The discovery of lithium in Reasi now offers a sliver of hope for the country to build self-sufficiency in this sector. Also, with more than 500 ppm (parts per million) quality, the find hints at a higher level of enrichment—more than 300 ppm is considered a good quality value.

Prof. Ashok Jhunjhunwala, a research scientist in electric vehicles who teaches at IIT-Madras, cautiously notes that it will be a long wait before further exploration can reveal a clearer picture. Yet, he thinks the discovery could quicken India's ambitious EV adoption programme. "It will enable us to make batteries for EVs with our own raw material," he says. The former advisor to the Union ministry of new and renewable energy was alluding to India's mission of expanding EVs to 40 per cent of two-wheelers and private cars, and almost 100 per cent of the commercial buses segment by 2030 to cut petroleum consumption by 156 MT, worth Rs 3.5 lakh crore. Jain says that even if only a portion of the find can be converted into lithium, it will not only help India meet its domestic demand but also be exported.